CENG381 Stochastic Processes

Stationary Distribution - Answer Key 1

Q1 Solution

a) Transition matrix:

From Bus: P(Bus follows Bus) = 1 - 3/5 = 2/5, P(Car follows Bus) = 3/5

From Car: P(Bus follows Car) = 1/4, P(Car follows Car) = 3/4

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            Bus    Car
Bus  [  2/5    3/5  ]
Car  [  1/4    3/4  ]
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b) Solve pi * P = pi:

pi_B = pi_B*(2/5) + pi_C*(1/4)

pi_C = pi_B*(3/5) + pi_C*(3/4)

pi_B + pi_C = 1

From the first equation:

pi_B - (2/5)*pi_B = (1/4)*pi_C

(3/5)*pi_B = (1/4)*pi_C

pi_C = (12/5)*pi_B

Substituting into pi_B + pi_C = 1:

pi_B + (12/5)*pi_B = 1

(17/5)*pi_B = 1

pi_B = 5/17 ~ 0.294

pi_C = 12/17 ~ 0.706

c) Interpretation:

pi_B = 5/17 ~ 29.4% of vehicles on the road are buses in the long run.

Q2 Solution

a) Verify pi * P = pi:

(pi*P)[E] = pi(E)*(1-p) + pi(W)*q

= [q/(p+q)]*(1-p) + [p/(p+q)]*q

= [q(1-p) + pq] / (p+q)

= [q - qp + pq] / (p+q)

= q/(p+q) = pi(E) (OK)

(pi*P)[W] = pi(E)*p + pi(W)*(1-q)

= [q/(p+q)]*p + [p/(p+q)]*(1-q)

= [qp + p(1-q)] / (p+q)

= [qp + p - pq] / (p+q)

= p/(p+q) = pi(W) (OK)

b) Recurrence and convergence:

M_t = M_{t-1} * P, so M_t = M_0 * P^t.

Define Delta_t = mu_t(E) - q/(p+q). Then:

Delta_{t+1} = (1-p-q) * Delta_t

So Delta_t = (1-p-q)^t * Delta_0.

Since 0 < p+q < 1 (given p,q>0 and p+q<1), we have |1-p-q| < 1,

so Delta_t -> 0, meaning M_t -> pi as t -> infinity.

c) For p=0.3, q=0.4:

pi(E) = 0.4/0.7 = 4/7 ~ 0.571

pi(W) = 0.3/0.7 = 3/7 ~ 0.429

lambda_2 = 1 - 0.3 - 0.4 = 0.3

Eigendecomposition: P = S^(-1) D S where D = diag(1, 0.3)

P^(2) entries:

P^(2)_{EE} = q/(p+q) + lambda_2^2 * p/(p+q)

= 4/7 + (0.09)*(3/7)

= 4/7 + 0.27/7

= (4 + 0.27)/7 ~ 0.610

P^(2)_{EW} = p/(p+q) - lambda_2^2 * p/(p+q)

= 3/7 - 0.09*(3/7) ~ 0.390

P^(2)_{WE} ~ 4/7 - 0.09*(4/7) ~ 0.571 - 0.051 = 0.520 [check: 0.7*0.4+0.3*0.4=0.28+0.12=0.40...]

Direct computation: P^(2)_{EE} = (0.7)^2 + (0.3)(0.4) = 0.49+0.12 = 0.61

P^(2)_{EW} = 0.7*0.3 + 0.3*0.6 = 0.21+0.18 = 0.39

P^(2)_{WE} = 0.4*0.7 + 0.6*0.4 = 0.28+0.24 = 0.52

P^(2)_{WW} = 0.4*0.3 + 0.6*0.6 = 0.12+0.36 = 0.48

Q3 Solution

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Stationary Distribution - Answer Key 1

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a) Solve pi * P = pi:

pi_0*(1/2) + pi_1*(1/4) + pi_2*(1/6) = pi_0 ... (i)

pi_0*(1/3) + pi_1*(1/2) + pi_2*(1/3) = pi_1 ... (ii)

pi_0*(1/6) + pi_1*(1/4) + pi_2*(1/2) = pi_2 ... (iii)

pi_0 + pi_1 + pi_2 = 1   ... (iv)

From (i): -pi_0/2 + pi_1/4 + pi_2/6 = 0 => -6*pi_0 + 3*pi_1 + 2*pi_2 = 0

From (iii): pi_0/6 + pi_1/4 - pi_2/2 = 0 => 2*pi_0 + 3*pi_1 - 6*pi_2 = 0

Notice the matrix is symmetric: P_{ij} = P_{ji}, so by detailed balance

pi_i * P_{ij} = pi_j * P_{ji} is satisfied when pi is uniform!

Check: if pi_0 = pi_1 = pi_2 = 1/3, then rows of P each sum to 1

and all columns of P each also sum to 1 (doubly stochastic).

=> pi = (1/3, 1/3, 1/3)

b) Verification:

(pi*P)[0] = (1/3)*(1/2) + (1/3)*(1/4) + (1/3)*(1/6)

          = (1/3)*(1/2 + 1/4 + 1/6)

          = (1/3)*(6/12 + 3/12 + 2/12)

          = (1/3)*(11/12) -- wait, let's recheck column sum:

Column 0 sum: 1/2 + 1/4 + 1/6 = 6/12+3/12+2/12 = 11/12 != 1

So P is NOT doubly stochastic. Solve properly:

From equations: pi_0 = pi_2 (by symmetry of equations (i) and (iii)).

Let pi_0 = pi_2 = x, pi_1 = y. Then 2x + y = 1.

From (i): x/2 + y/4 + x/6 = x => (3x+6y/4+2x)/6 is wrong; redo:

  -x/2 + y/4 + x/6 = 0 => multiply by 12: -6x + 3y + 2x = 0

  => 3y = 4x => y = 4x/3

  2x + 4x/3 = 1 => 10x/3 = 1 => x = 3/10

  y = 4/10 = 2/5

pi = (3/10, 2/5, 3/10) = (0.30, 0.40, 0.30)

c) Detailed balance check:

pi_0 * P_{01} = 0.30 * (1/3) = 0.10

pi_1 * P_{10} = 0.40 * (1/4) = 0.10 (OK)

pi_0 * P_{02} = 0.30 * (1/6) = 0.05
pi_2 * P_{20} = 0.30 * (1/6) = 0.05 (OK)

pi_1 * P_{12} = 0.40 * (1/4) = 0.10
pi_2 * P_{21} = 0.30 * (1/3) = 0.10 (OK)

Detailed balance holds => the chain is time-reversible.
This means the long-run rate of flow from i to j equals the rate
from j to i: the process looks the same forwards and backwards.